

Background

Motivation:

- Wetlands are biologically rich ecosystems that support diverse communities across multiple taxonomic groups
- Wetland restoration is increasingly used as a management strategy, but long-term biodiversity outcomes remain uncertain.



Figure 1: Field data collection in a wetland site during biodiversity sampling.

- Understanding whether restored wetlands function similarly to natural systems is critical for effective conservation planning.

- Evaluating species richness and conservation-relevant species helps assess ecological recovery beyond simple habitat restoration.

Research objectives:

- Compare overall and taxa-specific biodiversity metrics between restored and natural wetlands to assess restoration effectiveness.
- Identify biodiversity hotspots and low-diversity sites to inform conservation planning and prioritize management efforts.
- Examine multivariate biodiversity patterns to uncover underlying ecological structure and gradients across wetland sites.

Methodology

Data : Biodiversity data from 57 restored and natural wetland sites (2009-2011), including richness and abundance of birds ,anurans ,fish ,snakes and turtles

Study Design : Comparative analysis of biodiversity metrics between restored and natural wetland sites

Analysis :Descriptive statistics and outlier detection were performed Group differences were assessed using t-tests and levene's test

Multivariate analysis :
Principal Component Analysis (PCA) was applied to standardized biodiversity variables to identify dominant ecological patterns.

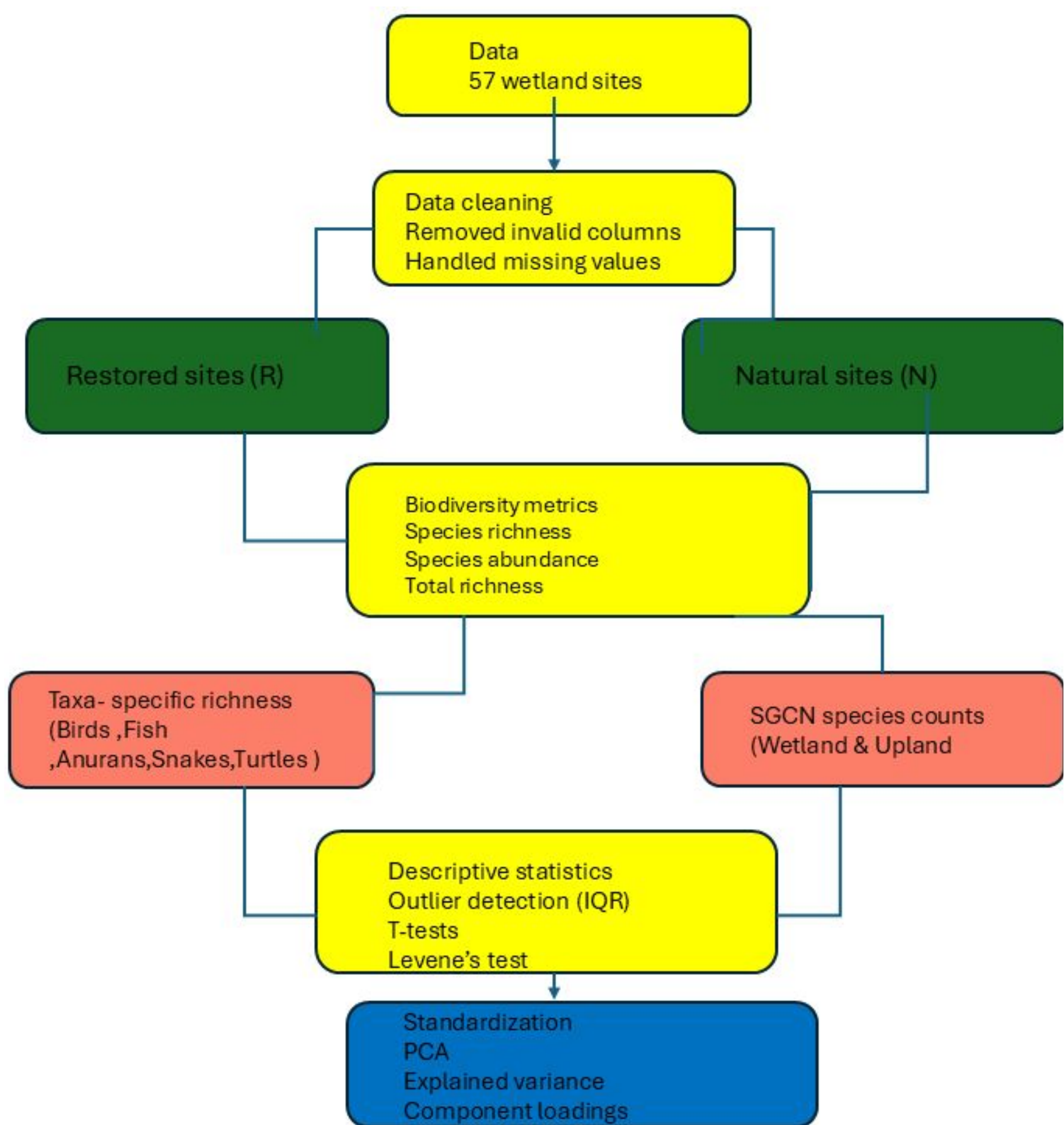


Figure 2: Flowchart of the full study

Results

Total Species Richness in Restored and Natural Wetland Sites

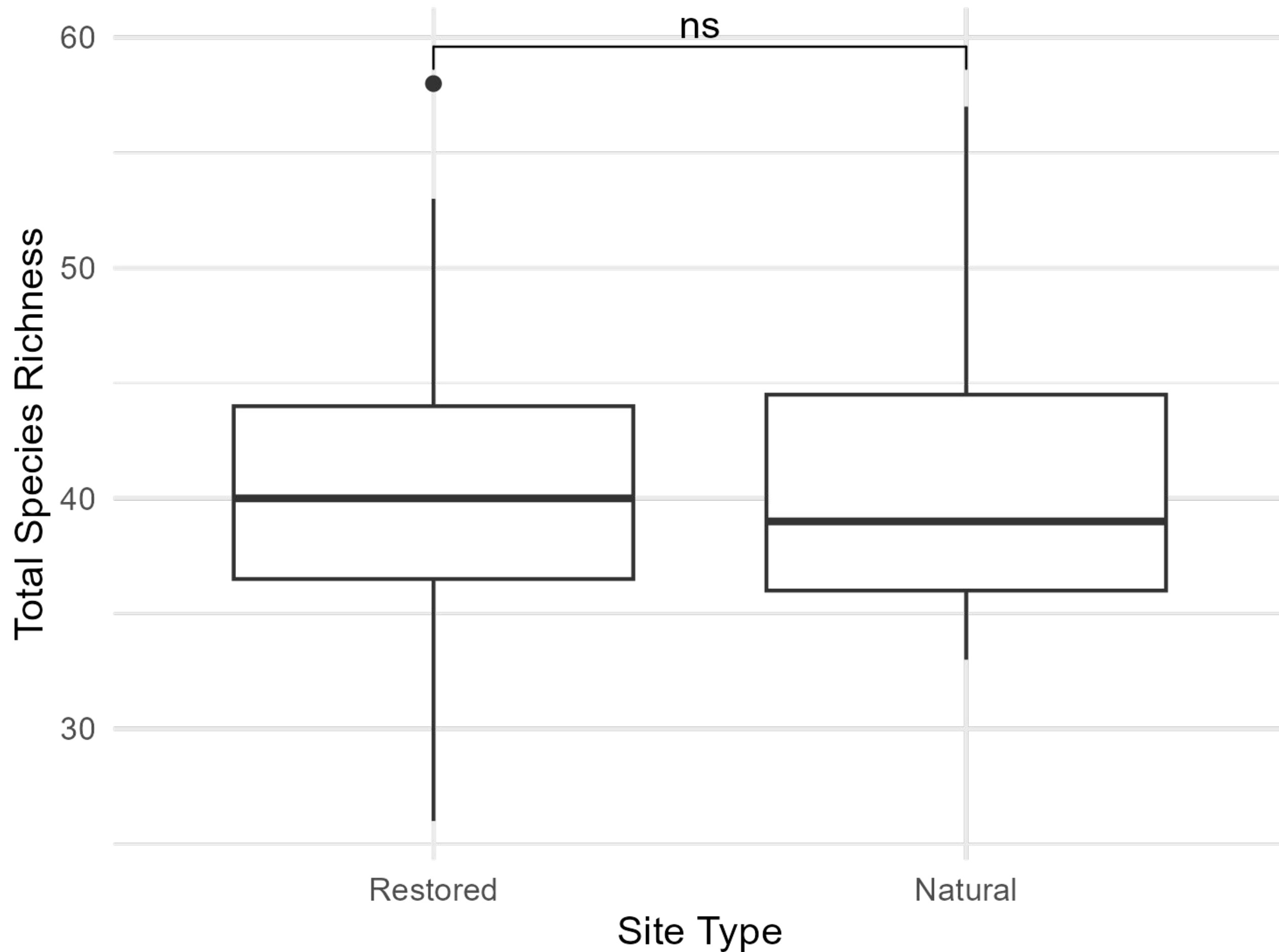


Figure 3:Boxplot comparing total species richness between restored and natural wetland sites.

- Total species richness did not differ significantly between restored and natural wetland sites.
- Median richness values and overall variability were comparable across site types, indicating that restored wetlands can support biodiversity levels like natural wetlands.

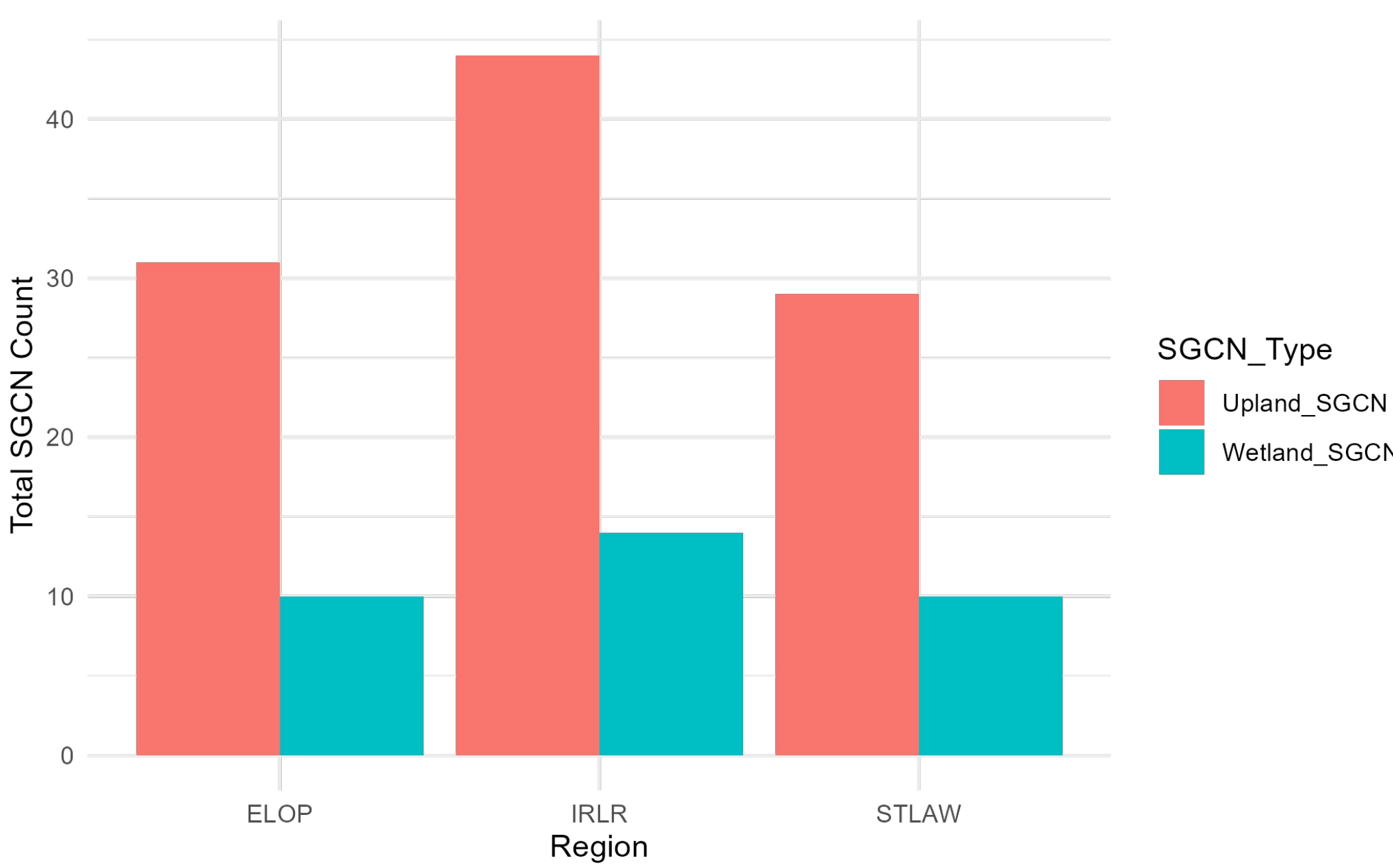


Figure 4: Taxa-Specific Species Richness in Restored and Natural Wetland Sites.

- Taxa-specific species richness patterns were generally similar between restored and natural sites, with modest variation observed among different taxonomic groups.
- Upland habitats consistently showed higher SGCN counts than wetland habitats across all regions, indicating greater species concentration in upland areas.

Results

Multivariate Analysis of Biodiversity Patterns

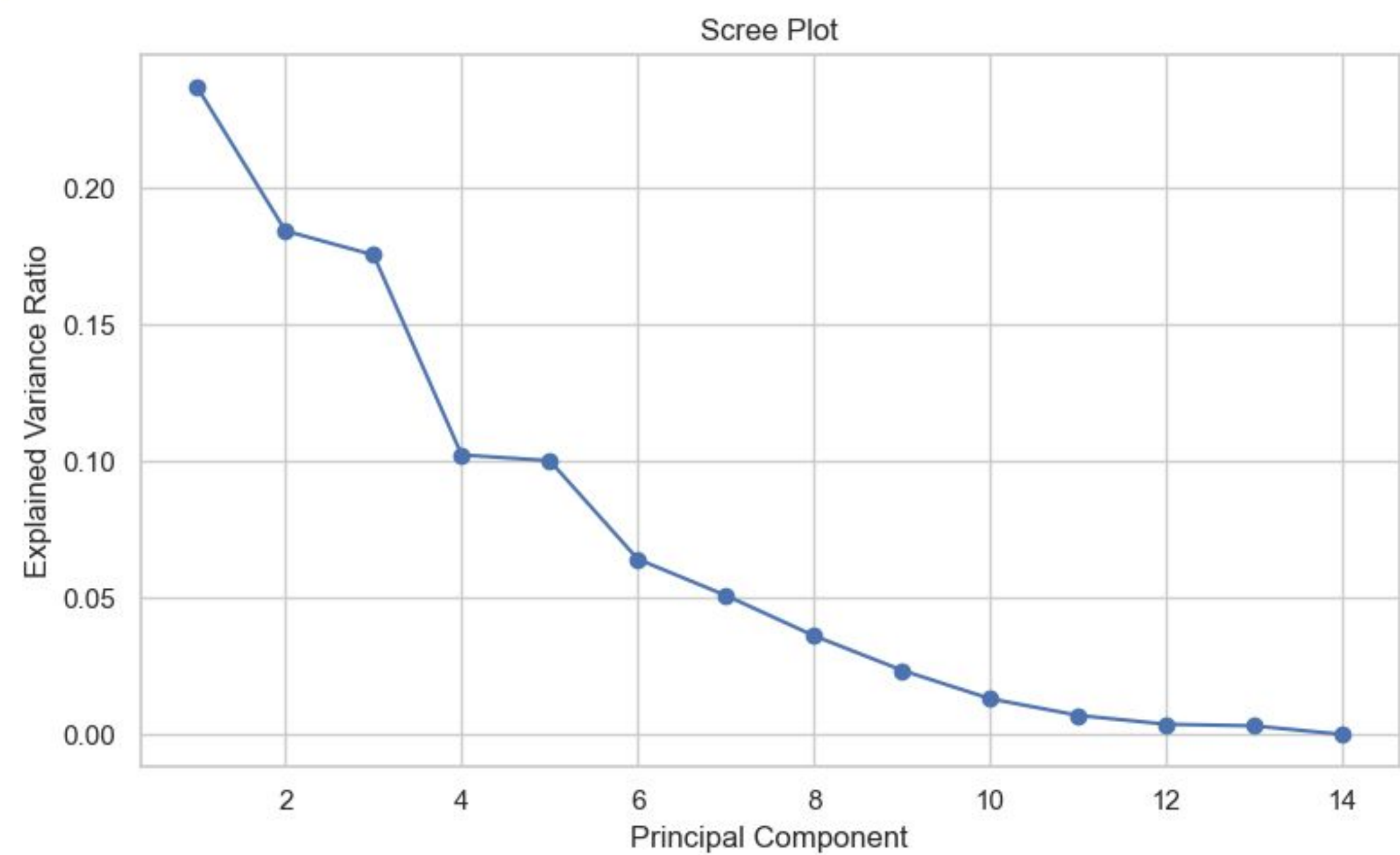


Figure 5: Explained Variance of Principal Components (Scree Plot)

Most variation is explained by the first few principal components, with decreasing contributions from later components.

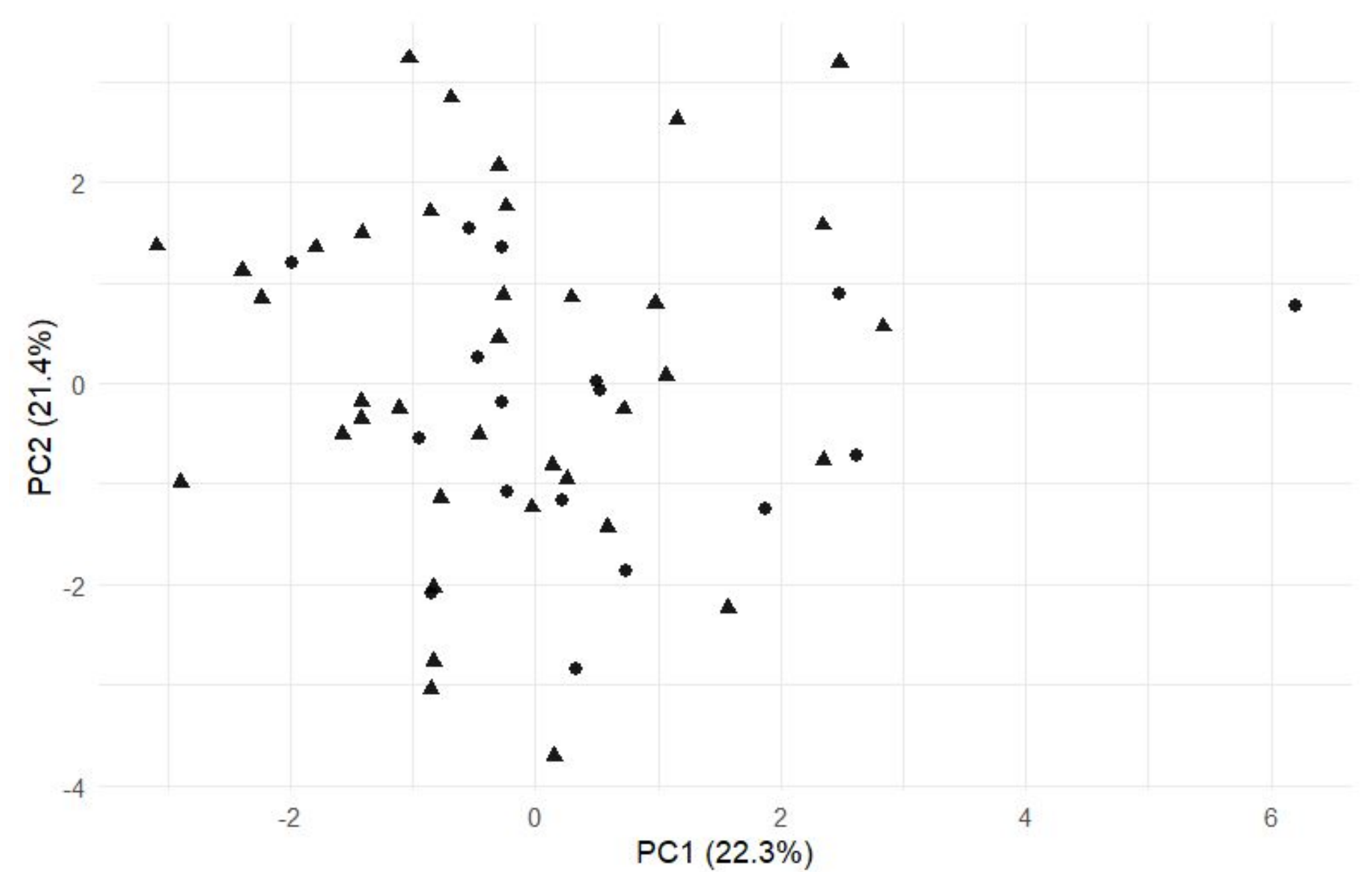


Figure 6 : Principal Component Analysis of Wetland Biodiversity Metrics

Biodiversity variation was distributed across multiple principal components, with no clear separation between restored and natural sites along PC1 and PC2.

This indicates that wetland biodiversity is influenced by complex, multi-taxa ecological gradients rather than restoration status alone.

Conclusion

- Restored wetlands supported overall biodiversity levels comparable to natural wetlands, indicating successful ecological recovery.
- Taxa-specific richness patterns were largely similar between restored and natural sites, suggesting consistent habitat functionality across groups.
- Multivariate analyses showed that biodiversity structure is influenced by multiple ecological gradients rather than restoration status alone.
- These findings support wetland restoration as an effective strategy for biodiversity conservation and ecosystem management.

References

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